Chapter 1: Statistical Learning Theory

Part III — Statistical Foundations of Deep Learning & Reinforcement Learning

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Abstract

Why should a model that fits the training data also fit data it has never seen? That question — generalisation — is the one statistical learning theory answers, and the answer is a concentration argument: the training error is an average of independent terms, so it concentrates around the true error, and the only thing that can break the link is a hypothesis class rich enough to chase the noise. This chapter builds the bound from Hoeffding's inequality to the finite-class guarantee, names the capacity measures (VC dimension, Rademacher complexity) that extend it to infinite classes, and reads the bias–variance trade-off and the modern surprise of double descent through the same lens. It is the spine the rest of Part III hangs on: every method that follows is an attempt to fit without overfitting.

Contents

1 Risk, empirical risk, and the gap between them 1

2 Concentration: why training error tracks true error 2

3 Infinite classes, and the bias–variance picture 2

1 Risk, empirical risk, and the gap between them

Fix a distribution $\mathcal{D}$ over pairs (x, y) and a loss ℓ. The object we actually care about is the **risk** — the expected loss of a hypothesis h on fresh data,

$$R(h) = \mathbb{E}_{(x,y)\sim\mathcal{D}}\big[\ell(h(x), y)\big],$$

but $\mathcal{D}$ is unknown, so we cannot evaluate it. What we can evaluate, on a sample $S = \{(x_i, y_i)\}_{i=1}^n$ drawn i.i.d. from $\mathcal{D}$, is the **empirical risk**

$$\widehat{R}_S(h) = \frac{1}{n}\sum_{i=1}^{n} \ell(h(x_i), y_i).$$

Empirical risk minimisation (ERM) returns the hypothesis $\hat{h} = \arg\min_{h\in\mathcal{H}} \widehat{R}_S(h)$ from a class $\mathcal{H}$. The whole game is the **generalisation gap** $R(\hat{h}) - \widehat{R}_S(\hat{h})$: training error is what we see, true error is what we want, and learning theory bounds the difference.

For a *single, fixed* h the gap is easy, because $\widehat{R}_S(h)$ is an average of n i.i.d. terms with mean $R(h)$. The law of large numbers says it converges; concentration says how fast.

2 Concentration: why training error tracks true error

Theorem 2.1 (Hoeffding's inequality)**.** Let $Z_1, \dots, Z_n$ be independent with $Z_i \in [a, b]$ and mean μ. Then for any $\varepsilon > 0$,

$$\Pr\left(\left|\tfrac{1}{n}\textstyle\sum_i Z_i - \mu\right| \geq \varepsilon\right) \;\leq\; 2\exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right).$$

With a bounded loss in $[0, 1]$ and $Z_i = \ell(h(x_i), y_i)$, this says the empirical risk of a fixed h is within ε of its true risk with probability at least $1 - 2e^{-2n\varepsilon^2}$ — the gap shrinks like $1/\sqrt{n}$. The middle panel of the figure shows exactly this: several running empirical means fold into the true mean, trapped inside the Hoeffding band $\pm\sqrt{\log(2/\delta)/(2n)}$.

The catch is ERM. It does not pick a fixed h; it picks the h that looks best *on this sample*, so $\widehat{R}_S(\hat{h})$ is a biased, optimistic estimate. The fix is to make the guarantee hold for every h in the class at once, with a union bound.

Theorem 2.2 (Finite-class generalisation bound)**.** Let $\mathcal{H}$ be finite and the loss lie in $[0, 1]$. With probability at least $1 - \delta$ over the sample, every $h \in \mathcal{H}$ satisfies

$$R(h) \;\leq\; \widehat{R}_S(h) + \sqrt{\frac{\log|\mathcal{H}| + \log(2/\delta)}{2n}}.$$

Proof. Fix h. By Hoeffding, $\Pr\left(R(h) - \widehat{R}_S(h) \geq \varepsilon\right) \leq e^{-2n\varepsilon^2}$. Apply the union bound over the $|\mathcal{H}|$ hypotheses: the probability that *any* one violates the gap is at most $|\mathcal{H}|\, e^{-2n\varepsilon^2}$. Set this equal to δ and solve for $\varepsilon = \sqrt{(\log|\mathcal{H}| + \log(1/\delta))/(2n)}$; the two-sided $\log(2/\delta)$ version follows by accounting for both tails. □

Read the bound. The gap grows with $\log|\mathcal{H}|$ — the richer the class, the more ways to get lucky on the training sample — and shrinks with n. That single inequality is the whole moral of the field: *generalisation = fit − capacity*$/\sqrt{n}$.

3 Infinite classes, and the bias–variance picture

$\log|\mathcal{H}|$ is useless when $\mathcal{H}$ is infinite — every linear classifier, every network. The repair is to measure *effective* size by how many label patterns the class can realise. The **VC dimension** $d_{\mathrm{VC}}(\mathcal{H})$ is the largest number of points the class can **shatter** (label in all 2^d possible ways); it replaces $\log|\mathcal{H}|$ in a bound of the same shape,

$$R(h) \;\lesssim\; \widehat{R}_S(h) + \sqrt{\frac{d_{\mathrm{VC}}\,\log(n/d_{\mathrm{VC}}) + \log(1/\delta)}{n}}.$$

Rademacher complexity is the tighter, data-dependent cousin: it measures how well the class can fit a vector of random ± 1 signs — literal noise — so a class that can fit noise generalises less. Both make the same point as the finite bound, with capacity now a property of the class's shape rather than its cardinality.

This is the statistical content of the **bias–variance trade-off**. A small class has high *bias* (it cannot represent the truth) but low *variance* (it cannot chase the noise); a large class is the reverse. Test error is their sum, so it falls, bottoms out, and rises — the U-shape in the left panel, where polynomial degree plays the role of capacity.

Example 3.1 (Double descent)**.** The classical U is not the end of the story. Push the capacity *past* the point where the model can interpolate the training set exactly ($p = n$

parameters) and test error, after spiking at the interpolation threshold, *falls again* — the right panel. At $p = n$ there is exactly one zero-training-error fit and it is wildly unstable; beyond it, many interpolating fits exist and gradient methods select the smallest-norm one, which is smooth and generalises. Classical theory describes the first descent; modern over-parameterised models live on the second. The bound is not wrong — the right capacity measure is the norm of the solution, not the raw parameter count.

Summary

Learning theory bounds the generalisation gap $R(h) - \widehat{R}_S(h)$. For a fixed h, Hoeffding gives an $O(1/\sqrt{n})$ gap; ERM needs the bound to hold uniformly over $\mathcal{H}$, which a union bound buys at the cost of a $\sqrt{\log|\mathcal{H}|/n}$ capacity term. For infinite classes, VC dimension and Rademacher complexity replace $\log|\mathcal{H}|$, giving $R(h) \lesssim \widehat{R}_S(h) + \sqrt{\text{capacity}/n}$. This is the bias–variance trade-off: too little capacity biases, too much chases noise, test error is U-shaped — except that past the interpolation threshold, over-parameterised models descend a second time, because the operative capacity is the norm of the solution, not the parameter count.